Naming groups in the periodic table

Elements on the periodic table can be divided into metals and non-metals. They can be further broken down into different groups based on their properties.

The metals found in Group 1 are very **reactive**. This means they react violently with water. They are also soft and can be cut with a knife. Examples include lithium, sodium and potassium, which are also known as the **alkali metals**. Hydrogen is in Group 1 of the periodic table but it is not considered part of the alkali metals because it is a **non-metal**.

The metals found between Groups 2 and 3 are called **transition metals**. They have the typical properties of metals – they are hard, dense and have high melting and boiling points. The **semi-metals** or **metalloids** are located along the stepped line between metals and non-metals. Boron, silicon, germanium, arsenic, antimony, tellurium and polonium are all semi-metals. They are called semi-metals because they have some characteristics of metals and some of non-metals. For example, silicon is shiny, but it is brittle, so it would break it you hit it with a hammer.

The non-metals in Group 7 are very **reactive**. They are called **halogens**. Halogens are poisonous non-metals. Examples include fluorine, chlorine, bromine, iodine and astatine. Like most non-metals, they have low melting and boiling points.

Group 8 – sometimes known as Group 0 – is found on the right-hand side of the periodic table. These elements are very special because they do not react to form compounds. They are found by themselves in nature. They are called **noble gases**.

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Groups in the periodic table

Use the Resource sheets 'How is the periodic table arranged?' and 'Naming groups in the periodic table' to answer the following questions.

☞ **1** On the periodic table on the 'How is the periodic table arranged?' Resource sheet, colour:

- B (boron), Si (silicon), Ge (germanium), As (arsenic), Sb (antimony), Te (tellurium) and Po (polonium) in green. These elements are called semi-metals, or metalloids.
- the alkali metals in yellow
- the transition metals in orange
- the halogens in blue
- the noble gases in pink
- all the remaining non-metals in purple. Remember, hydrogen is a non-metal.

2 Name the elements in Group1. _____

_____

3 What is unusual about the Group 1 elements? _____

4 Which group is Br (bromine) in? What do we call this group? _____

5 The metals between Groups 2 and 3 are called transition metals. How many transition metals are there? _____

6 Which of these elements is the odd one out? Circle it. Li, Na, Ca, K, Rb, Cs

7 Match the sentence halves to summarise the different groups.

Alkali metals react	do not react to form compounds.
Group 7 elements, also known as	violently in water.
Noble gases, found in Group 8 (or 0),	and boiling points and are dense.
Transition metals have high melting	halogens, are very reactive poisonous non-metals.

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